

Worked example. Let $x = [1, 2]$ and $y = [2, 1]$. Then $x \cdot y = 1 \cdot 2 + 2 \cdot 1 = 4$, $\|x\| = \|y\| = \sqrt{5}$, so $\cos(\theta) = 4/5 = 0.8$. The vectors are similar but not identical in direction — check: they are mirror images across the diagonal.

Matrices: functions, not tables

A **matrix** A of shape $m \times n$ is best understood not as a grid of numbers but as a *linear function* that maps vectors from $\mathbb{R}^n$ to $\mathbb{R}^m$ via $v \mapsto Av$. Every linear layer in a neural network is exactly this: `output = W @ input + b`. Matrix multiplication is function composition — $B(Av) = (BA)v$ — which is why a stack of linear layers *without nonlinearities* collapses into one linear layer: the product W_2W_1 is just another matrix. This single fact is the reason activation functions exist, and it is a classic interview question.

Key facts you should be able to state cold: matrix multiplication requires inner dimensions to match ($m \times n$ times $n \times p$ gives $m \times p$); it is associative but **not commutative** ($AB \neq BA$ in general); the transpose reverses order, $(AB)^T = B^T A^T$; and the inverse A^{-1} exists only for square matrices with nonzero determinant (equivalently: full rank, no zero eigenvalues, columns linearly independent — interviewers love asking for all the equivalent conditions).

Rank is the number of linearly independent columns (equivalently rows) — the true dimensionality of the space the matrix can reach. A rank-deficient design matrix X means some feature is a linear combination of others (perfect multicollinearity), and $X^T X$ becomes non-invertible — which is precisely why the normal equation for linear regression can fail and why Ridge regression adds λI to fix it. Chapter 5 builds on this.

Eigenvalues and eigenvectors: the directions a matrix cannot rotate

For a square matrix A , an **eigenvector** v is a special direction that A does not rotate — it only stretches or shrinks it:

$$Av = \lambda v$$

The stretch factor λ is the **eigenvalue**. Think of a matrix as a transformation of space: most directions get rotated somewhere else, but eigenvectors are the axes of the transformation, and eigenvalues tell you how much each axis is scaled.

Worked example. Let A be the diagonal matrix with entries 2 and 3 (rows (2, 0) and (0, 3)). Any vector along the x-axis satisfies $Av = 2v$, any vector along the y-axis satisfies $Av = 3v$. Eigenvalues are 2 and 3, eigenvectors are $[1, 0]$ and $[0, 1]$. For non-diagonal matrices the eigenvectors are not axis-aligned, but the idea is identical. To find eigenvalues by hand, solve the characteristic equation $\det(A - \lambda I) = 0$: for the matrix with rows (2, 1) and (1, 2), this is $(2 - \lambda)^2 - 1 = 0$, so $\lambda = 1$ or $\lambda = 3$ (with eigenvectors along $[1, -1]$ and $[1, 1]$ — check by multiplying).

Why ML interviews care:

Compute how surprising the observed data would be under that assumption. If the data is too surprising, reject H_0 .

The **p-value** is the probability, *assuming H_0 is true*, of observing data at least as extreme as what you saw. It is **not** the probability that H_0 is true — that misstatement is the single most common statistics error in interviews, and interviewers bait it deliberately. A p-value below a pre-chosen threshold α (conventionally 0.05) means "reject H_0 "; α is exactly the **Type I error** rate — the probability of rejecting a true null (false alarm). **Type II error** (β) is failing to reject a false null (missed detection), and **power** $= 1 - \beta$ is the probability of catching a real effect. Power grows with effect size and sample size; that is why you run a *power analysis before* an experiment to decide how much data you need.

The t-test compares means. The one-sample statistic is $t = (\bar{x} - \mu_0)/(s/\sqrt{n})$: how many standard errors is the sample mean from the hypothesized mean? The two-sample version compares two groups (e.g., treatment vs control conversion values). It assumes approximately normal sample means (CLT gives you this for large n) and, in the classic form, similar variances (Welch's variant relaxes this — say "Welch's t-test" in interviews when unsure).

The chi-square test handles categorical counts: $\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$ over cells of a contingency table, where O is observed and E is the count expected under independence. Use it for "did signup rate differ across these five landing pages?" or goodness-of-fit checks.

Confidence intervals. A 95% CI for a mean is $\bar{x} \pm 1.96 \cdot s/\sqrt{n}$. Precise meaning (another bait question): if you repeated the experiment many times, 95% of the intervals constructed this way would contain the true value. It is a statement about *the procedure*, not "there's a 95% chance the truth is in this particular interval."

A/B testing packages all of this: randomize users into control (A) and treatment (B), pre-register the metric and α , run the power analysis, collect data, run the test (t-test for continuous metrics, chi-square or two-proportion z-test for conversion rates). Practical failure modes interviewers probe: **peeking** (checking repeatedly and stopping when significant inflates false positives far above α); **multiple comparisons** (testing 20 metrics at $\alpha = 0.05$ expects one false positive — use Bonferroni or FDR corrections); **novelty effects** (users click new things briefly); and **interference** (one user's treatment affecting another's behavior, e.g., in marketplaces or social feeds). Chapter 27 revisits A/B testing for model deployment.

Maximum likelihood and MAP

MLE: let the data speak

You have data and a model with unknown parameters θ . **Maximum Likelihood Estimation** picks the θ that makes the observed data most probable:

The gradient line is the chain rule from this chapter applied to $\mathcal{L}(w) = \frac{1}{n} \|y - Xw\|^2$. Note `np.linalg.solve` rather than explicitly inverting $X^T X$ — inverting is slower and numerically worse, and saying so is an easy interview point. Both estimates recover `true_w` up to noise, and they agree to three decimals: the iterative and closed-form solutions of the same convex problem.

MLE for a coin, numerically

```
import numpy as np
flips = np.array([1,1,1,0,1,1,0,1,0,1]) # 7 heads out of 10
ps = np.linspace(0.01, 0.99, 99)
loglik = flips.sum()*np.log(ps) + (len(flips)-flips.sum())*np.log(1-ps)
print("MLE p-hat:", np.round(ps[np.argmax(loglik)], 2))
```

Output:

```
MLE p-hat: 0.7
```

The grid search lands exactly on the calculus answer 7/10 from the worked example — a useful sanity-check pattern: when you derive an estimator by hand, verify it numerically.

Entropy, cross-entropy, and KL divergence from scratch

```
import numpy as np
def entropy(p):
    p = np.asarray(p); return -np.sum(p * np.log2(p, where=p>0))
def cross_entropy(p, q):
    p, q = np.asarray(p), np.asarray(q); return -np.sum(p*np.log2(q, where=q>0))
def kl(p, q):
    return cross_entropy(p, q) - entropy(p)

P = [0.5, 0.5]; Q = [0.9, 0.1]
print("H(P) =", entropy(P), "bits | H(P,Q) =", round(cross_entropy(P,Q),3),
      "| KL(P||Q) =", round(kl(P,Q),3))
print("KL(Q||P) =", round(kl(Q,P),3), " (asymmetric!)")
```

Output:

```
H(P) = 1.0 bits | H(P,Q) = 1.737 | KL(P||Q) = 0.737
KL(Q||P) = 0.531 (asymmetric!)
```

The `where=p>0` guard implements the convention $0 \log 0 = 0$ without NaNs. The two KL values differ — the asymmetry stated in the theory section, demonstrated in four lines.

An A/B test from scratch — including its failure

```
import numpy as np
from math import erf, sqrt
rng = np.random.default_rng(7)
nA, nB = 10_000, 10_000
conv_A = rng.binomial(1, 0.100, nA) # control: 10.0% true rate
```

Statistical reasoning

Q16. Define p-value precisely. Your A/B test gives $p = 0.03$ — what can and can't you say?

The p-value is the probability, computed assuming the null hypothesis is true, of data at least as extreme as observed. With $p = 0.03$ you can say: "if there were truly no effect, results this extreme would occur 3% of the time; at $\alpha = 0.05$ we reject the null." You cannot say: there's a 97% chance the treatment works, or the effect is large, or the finding will replicate. Effect size comes from the estimate and CI, not the p-value.

Q17. What are Type I and Type II errors, and how do they trade off?

Type I: rejecting a true null (false positive), rate α . Type II: failing to reject a false null (missed effect), rate β ; power = $1 - \beta$. Lowering α (stricter threshold) raises β for fixed data. You escape the tradeoff with more samples or bigger effects. Which error is worse is a product question: flagging fraud (Type I annoys users) vs missing fraud (Type II loses money).

Q18. Your A/B test shows no significant difference. The PM says "so the feature has no effect." How do you respond?

Non-significance is not evidence of no effect — the test may be under-powered. Check: (1) the confidence interval — if it includes effect sizes that would matter for the business, the test is inconclusive, not negative; (2) the pre-experiment power analysis — did we have the sample size to detect the minimum effect we care about? (3) data quality — randomization balance, metric definition. The chapter's Listing 5 shows a genuinely better treatment failing to reach significance at $n=10,000/\text{arm}$.